

$$24 \quad F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} = \frac{Q^2}{4\pi\epsilon_0 r^2}$$

Since each point charge is increased by adding $2Q$, they now have a charge of $3Q$ each.

$$F' = \frac{(3Q)(3Q)}{4\pi\epsilon_0 (3r)^2} = \frac{Q^2}{4\pi\epsilon_0 r^2} = F$$

Answer: C

25. Let the length of each grid be D .